

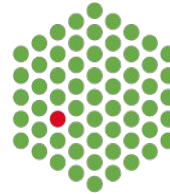
Practical: deconvolution of the clonality of a tumour using single-cell transcriptomic data

Martha Estefania Vázquez Cruz
International Laboratory for Human Genome Research
National Autonomous University of Mexico (LIIGH UNAM)
evazquez@liigh.unam.mx

Gabriel Pereira
Federal University of Health Sciences of Porto Alegre (UFCSPA)
Immunotherapy LAB (LAIT)
gabriel.pereira@ufcspa.edu.br



EMBL-EBI



Overview

- 1 Introduction
- 2 Approaches and considerations
- 3 Benchmark studies
- 4 Activity 1: Individual exercises (Bisque, BayesPrism, CIBERSORTx)
- 5 Activity 2: Final quiz

Estef Vazquez

Institution: LIIGH UNAM - National Autonomous University of Mexico
Cancer Genetics and Bioinformatics (CGBioLab)
Led by Professor Daniela Robles-Espinoza, PhD

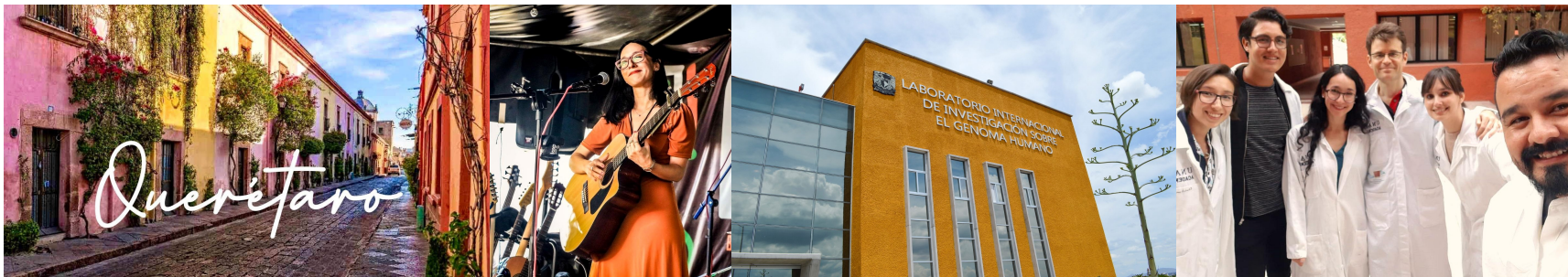


I am currently investigating the components of the tumor immune microenvironment of acral melanoma. My research interests include gaining a deeper understanding of immune evasion mechanisms in cancer, and leveraging multi-omics data integration to gain mechanistic insights. I hope my research contributes to improving cancer patient outcomes and to the development of more effective treatments globally.

[@estefvazquez.bsky.social](https://bsky.app/profile/estefvazquez.bsky.social)

 [@EstefVazque](https://twitter.com/EstefVazque)

Hobbies: Music, play guitar and sing, running, play with my crazy cat



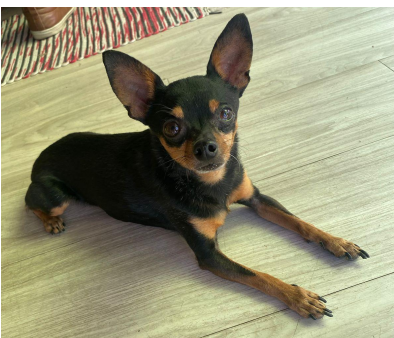
Gabriel Pereira



Institution: UFCSPA - Federal University of Health Sciences of Porto Alegre (Brazil)
Immunotherapy Lab (LAIT)
Led by Professor Cristina Bonorino, PhD

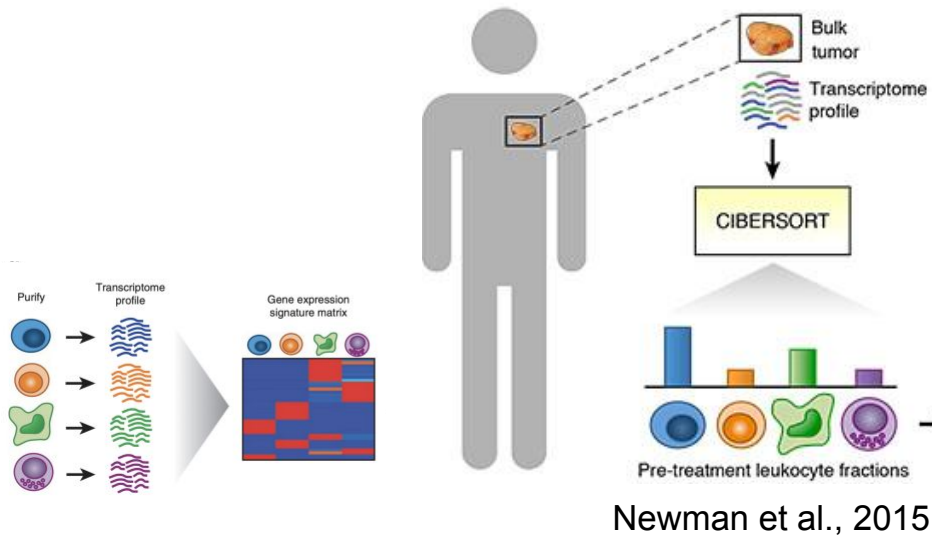
I am working on creating a single-cell and spatially resolved atlas of pancreatic cancer. My goal is to understand the tumor ecosystem and the spatial distribution of immune cells, and how this information can provide critical insights into prognosis and therapy response.

Hobbies: Play football, play with my dog and watching F1



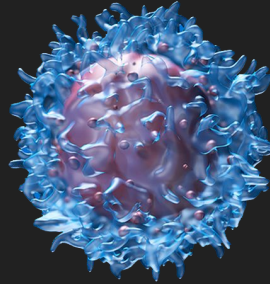
Deconvolution

Computational approaches to estimate cell-type composition from bulk RNA-seq.



Using scRNA-seq profiles to define **cell-type-specific signatures**

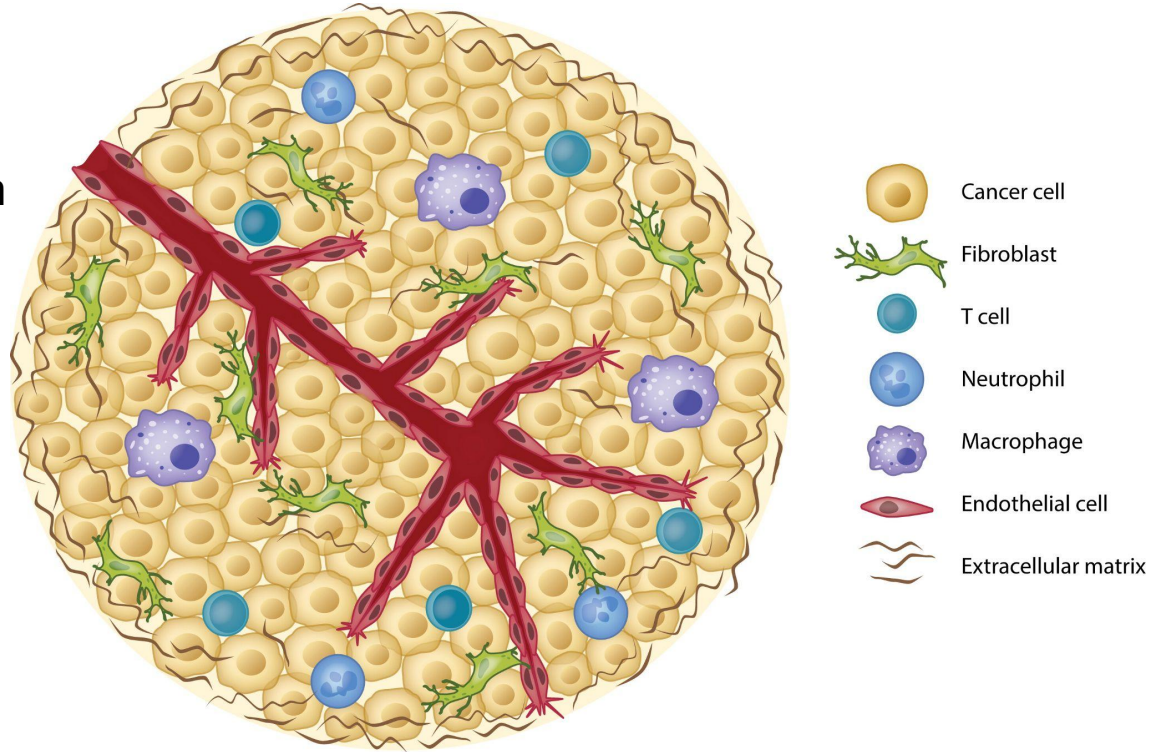
High resolution of single-cell data



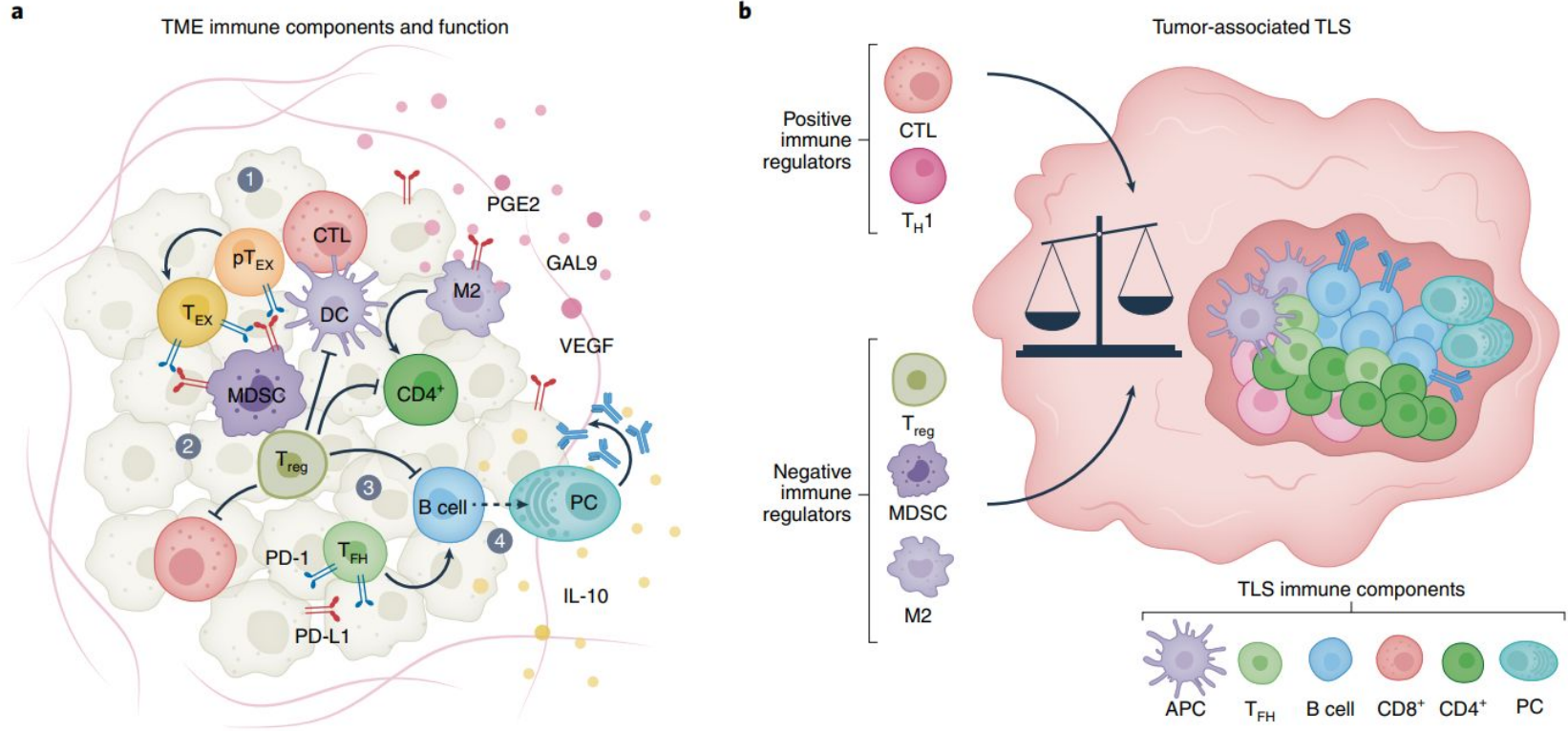
Cost-effectiveness of bulk RNA-seq

The tumour microenvironment impacts therapy response

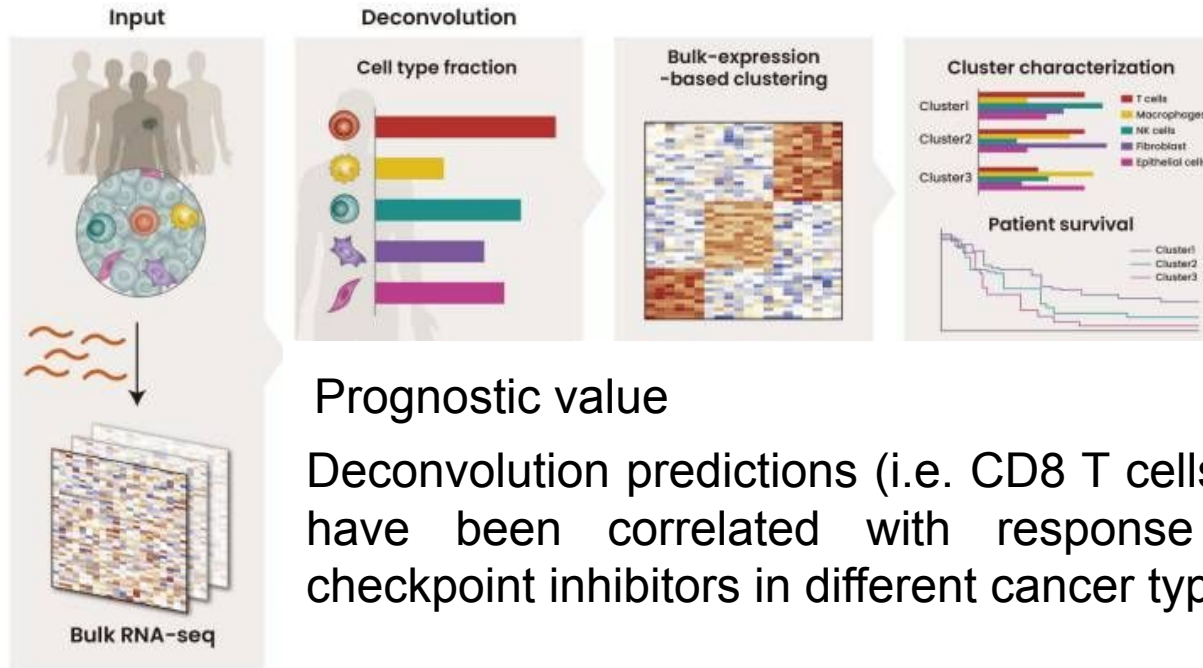
- Constant communication between tumour and immune cells
- Composition and activity of immune and stromal cell types.
- Clinical relevance



The tumour immune microenvironment in melanoma



Deconvolution to study the TME



Benchmarking studies

Sturm et al., 2019

- No 'one-size-fits-all' method.
- EPIC and quanTIseq as general purpose methods.

Finotello et al., 2024

- Impact of sources of variability: single-cell reference size, cellular resolution, cell-type specific mRNA bias, unknown cellular content, single-cell technology and tissue context.
- DWLS and Scaden for single-cell-informed deconvolution.

Huuki-Myers et al., 2025

- Mean Ratio in DeconvoBuddies
- Impact of library prep (polyA vs RiboZero)
- Post-mortem brain tissue
- Top performers: Bisque, hspe, CIBERSORTx

Cobos et al., 2020

- Impact of data normalization strategies on deconvolution performance.

White et al., 2024

- Levels of most major immune and stromal lineages were well predicted.
- High variability for sub-populations.
- CIBERSORTx

First and second generation methods

CIBERSORT
EPIC
xCell
quantiSeq
TIMER
MCP-counter

Pre-computed expression
signatures cover limited cell types.

CIBERSORTx
BayesPrism
MuSic
MuSic2
Bisque
Scaden (deep learning)

Provide more flexibility.
Use of scRNA-seq to build custom signatures.

Mathematical approaches and marker gene selection strategies

Method	Citation	Approach	Marker Gene Selection	Availability	Top Benchmark Performance
DWLS (Dampened weighted least-squares)	Tsoucas et al, Nature Comm, 2019 [5]	weighted least squares	-	R package on CRAN	Cobos et al. [18]
Bisque	Jew et al, Nature Comm, 2020 [6]	Bias correction: Assay	-	R package on GitHub	Dai et al. [17]
MuSiC (Multi-subject Single-cell)	Wang et al, Nature Communications, 2019 [7]	Bias correction: Source	Weights Genes	R package GitHub	Jin et al. [20]
BayesPrism	Chu et al., Nature Cancer, 2022 [8]	Bayesian	Pairwise t-test	Webtool R package on GitHub	Hippen et al. [22]
hspe (dtangle) (hybrid-scale proportion estimation)	Hunt and Gagnon-Bartsch, Ann. Appl. Stat. 2021 [9, 44]	High collinearity adjustment	Multiple options- default "ratio" 1vALL mean expression ratio	R package on GitHub	Dai et al. [17]
CIBERSORTx	Newman et al., Nat Biotech, 2019 [11]	Machine Learning	Differential Gene expression	Webtool, Docker Image	Jin et al. [20]

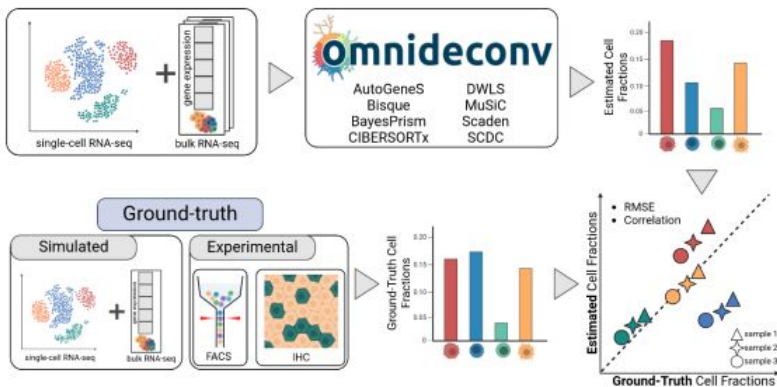
Huuki-Myers et al., 2023
Pre-print

Omnideconv

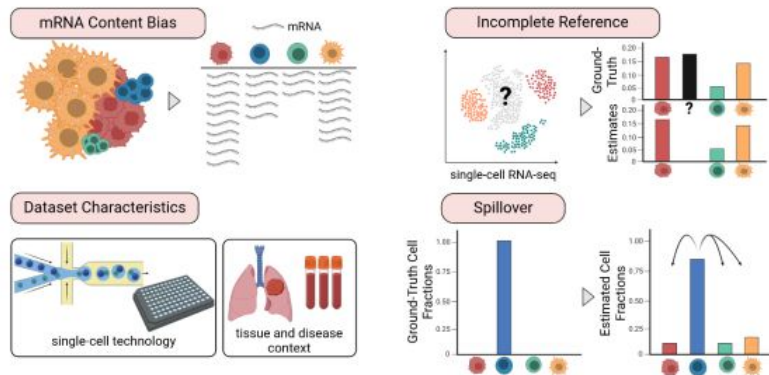
A) Omnideconv ecosystem



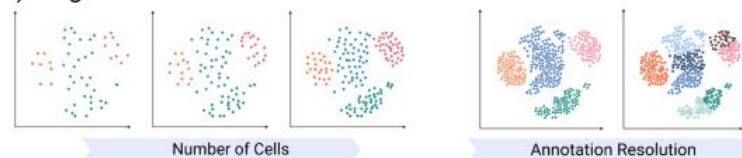
B) Benchmarking framework



C) Challenges addressed



D) Single-cell reference characteristics



Streamline deconvolution analysis and benchmarking

<https://omnideconv.org/>

<https://exbio.wzw.tum.de/deconvexplorer/>

OMNIDECONV

MENU

ENABLING THE DECONVOLUTION OF ANY CELL TYPE, TISSUE, AND ORGANISM

WELCOME TO OMNIDECONV!

omnideconv is an ecosystem of user-friendly tools and resources for cell-type deconvolution.

The omnideconv ecosystem has been developed in a joint effort of the [Group of Computational Biomedicine](#) (University of Innsbruck, Austria) and the [Group of Data Science in Systems Biology](#) (TUM School of Life Sciences, Germany).



Deconvolution DREAM Challenge

- Community assessment of deconvolution methods
- 28 community-contributed methods
- Estimated cell fractions vs ground truth
- Coarse vs fine grained subchallenges
- Challenges across methods
- Specificity, sensitivity, LOD, computational efficiency
- Per cell-type top performers

<https://dreamchallenges.org/>

Article | [Open access](#) | Published: 27 August 2024

Community assessment of methods to deconvolve cellular composition from bulk gene expression

[Brian S. White](#), [Aurélien de Reyniès](#), [Aaron M. Newman](#), [Joshua J. Waterfall](#), [Andrew Lamb](#), [Florent Petitprez](#), [Yating Lin](#), [Rongshan Yu](#), [Martin E. Guerrero-Gimenez](#), [Sergii Domanskyi](#), [Gianni Monaco](#), [Verena Chung](#), [Jineta Banerjee](#), [Daniel Derrick](#), [Alberto Valdeolivas](#), [Haojun Li](#), [Xu Xiao](#), [Shun Wang](#), [Frank Zheng](#), [Wenxian Yang](#), [Carlos A. Catania](#), [Benjamin J. Lang](#), [Thomas J. Bertus](#), [Carlo Piermarocchi](#), [Tumor Deconvolution DREAM Challenge consortium](#), ... [Andrew J. Gentles](#) 

[+ Show authors](#)

[Nature Communications](#) **15**, Article number: 7362 (2024) | [Cite this article](#)

<https://www.nature.com/articles/s41467-024-50618-0>

Mouse transcriptomics deconvolution

Bioinformatics Advances, 2024, 00, vbae032
<https://doi.org/10.1093/bioadv/vbae032>
 Advance Access Publication Date: 28 February 2024
 Application Note

OXFORD

Immunoinformatics

Making mouse transcriptomics deconvolution accessible with immunedeconv

Lorenzo Merotto¹, Gregor Sturm^{2,3}, Alexander Dietrich⁴, Markus List^{1,4},
 Francesca Finotello^{1,*}

¹Department of Molecular Biology, Digital Science Center (DISC), University of Innsbruck, Innsbruck 6020, Austria

²Biocenter, Institute of Bioinformatics, Medical University of Innsbruck, Innsbruck 6020, Austria

³Boehringer Ingelheim International Pharma GmbH & Co KG, Biberach 88400, Germany

⁴Data Science in Systems Biology, TUM School of Life Sciences, Technical University of Munich, Freising 85354, Germany

*Corresponding author: Institute of Molecular Biology, University of Innsbruck, Technikerstraße 25, 6020 Innsbruck, Austria; Digital Science Center (DISC), University of Innsbruck, Innrain 15, 6020 Innsbruck, Austria. E-mail: francesca.finotello@uibk.ac.at

Associate Editor: Guoqiang Yu

Abstract

Summary: Transcriptome deconvolution has emerged as a reliable technique to estimate cell-type abundances from bulk RNA sequencing data. Unlike their human equivalents, methods to quantify the cellular composition of complex tissues from murine transcriptomics are sparse and sometimes not easy to use. We extended the immunedeconv R package to facilitate the deconvolution of mouse transcriptomics, enabling the quantification of murine immune-cell types using 13 different methods. Through immunedeconv, we further offer the possibility of tweaking cell signatures used by deconvolution methods, providing custom annotations tailored for specific cell types and tissues. These developments strongly facilitate the study of the immune-cell composition of mouse models and further open new avenues in the investigation of the cellular composition of other tissues and organisms.

Availability and implementation: The R package and the documentation are available at <https://github.com/omnideconv/immunedeconv>.

1 Introduction

Bulk RNA sequencing (RNA-seq) captures the expression pro-

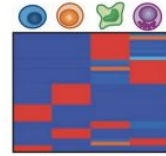
2 Extension of immunedeconv to the analysis of mouse transcriptomics

	Method	Output	Cell Types	Custom signature
Mouse-based methods 	mMcp counter		12 immune 4 non-immune	
	seqimmucc		10 immune	✓
	DCQ		17 immune stromal and stem cells	
	BASE		15 immune stromal and stem cells	
Human-based methods   	quanTiseq		10 immune Unknown content	
	EPIC		6 immune 2 non-immune Unknown content	✓
	CIBERSORT		22 immune	✓
	TIMER		6 immune	
	xCell		64	
	MCP counter		8 immune 2 non-immune	
	ABIS		17 immune	
	ConsensusTME		15 immune 2 non-immune	✓
	Estimate		Immune, stromal and tumor score Tumor purity	
		 : cell-type scores  : cell-type fractions		

Factors that may influence the performance of deconvolution

Cell-type specific markers

- Genes that are able to distinguish cell types with high specificity
- There is a need to refine cell type markers
- Single-cell characterization of rare cell types



Granularity

Coarse-grained vs fine-grained



Broad cell types



Finer subsets and states
(i.e. T-cell subsets)

Pre-processing of input data

- Normalization corrects for differences in the library size
- Logarithmic transformation of the input matrices can result in worse performance with the exception of BisqueRNA and bseq-sc
- Keeping input data in linear scale aids in assessing the cell proportions accurately

Avila Cobos et al. 2020; Jew et al. 2020, Jin & Liu 2021.

Other considerations

- Batch effect (i.e. variation between sc reference and bulk).
- Deconvolution performance differs by cell type
- Specificity (i.e. 'spillover')
- Assumptions
- Computational efficiency
- Underrepresentation of a cell type in the sc reference
- Tissue and disease context
- Robust and accurate estimations of cellular composition in a tumor, particularly when a cell type is well characterized.

The three pieces we need to deconvolve RNA-seq

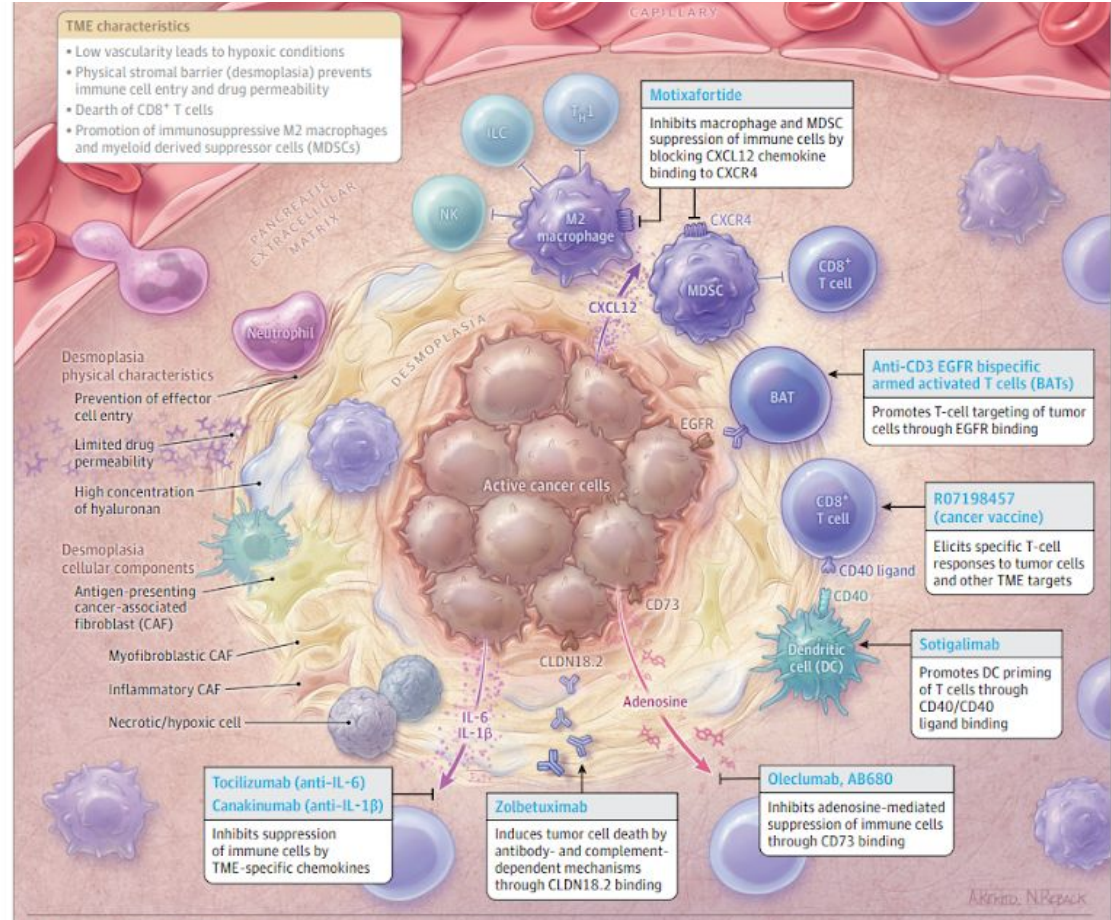
1. Bulk gene expression matrix
2. Single-cell reference
3. A deconvolution method

Hands-on

For this practical, we will work with Pancreatic Ductal Adenocarcinoma (PDAC) bulk data from the TCGA project (downsampled) and a PDAC scRNA-seq reference (Steele et al., 2020).

Pancreatic cancer (PDAC)

- Highly aggressive disease
- Desmoplastic reaction
- “Cold” tumor



Connect to your VM

1. Copy the `sc_deconvolution` folder from Penelope Cloud or clone the github repository
2. Locate the input files
3. Open Rstudio and the '`sc_deconvolution.Rmd`' file
4. Set your working directory

Note: All required libraries and input data are pre-installed on your virtual machine. The code and input data is also available in the GitHub repository: <https://github.com/estefvazquez/scDeconvolution> 

Load libraries

```
Library(BayesPrism)  
Library(BisqueRNA)  
Library(Biobase)  
Library(dplyr)  
Library(tidyR)
```


Load input data

1. `bulkdata <- tcga_pdac_counts.rds`
2. `bulk_tpms <- tcga_pdac_tpms.rds`
3. `clindata <- tcga_pdac_clindata.rds`
4. `sobj <- sc_ref_pdac_seurat.rds`

Seurat object

Exercise 1

Run Bisque

1. Convert bulk RNA-seq from a matrix to an ExpressionSet

```
bulk.eset <- Biobase::ExpressionSet(assayData = bulk.matrix)
```

2. Run deconvolution
3. Explore output

Execute the code interactively, line by line, to understand what each command does. Once you have completed running all the chunks, you can generate an HTML report by knitting the document. Got questions? We are here to help!

Exercise 2

Run BayesPrism

1. Prepare data
2. Run deconvolution
3. Explore output

Execute the code interactively, line by line, to understand what each command does. Once you have completed running all the chunks, you can generate an HTML report by knitting the document. Got questions? We are here to help!

Exercise 3

Run CIBERSORTx

Access your CIBERSORTx account: <https://cibersortx.stanford.edu/>

For this demonstration we will use the LM22 signature (Newman et al., 2015).

CIBERSORTx

Home

About

Contact

Register

Email

Password

Sign in

☐ Remember Me

User data generated before 8/15/23 currently not available.

CIBERSORTx

CIBERSORT

CIBERSORTx is an analytical tool from the [Alizadeh Lab](#) and [Newman Lab](#) to impute gene expression profiles and provide an estimation of the abundances of member cell types in a mixed cell population, using gene expression data.

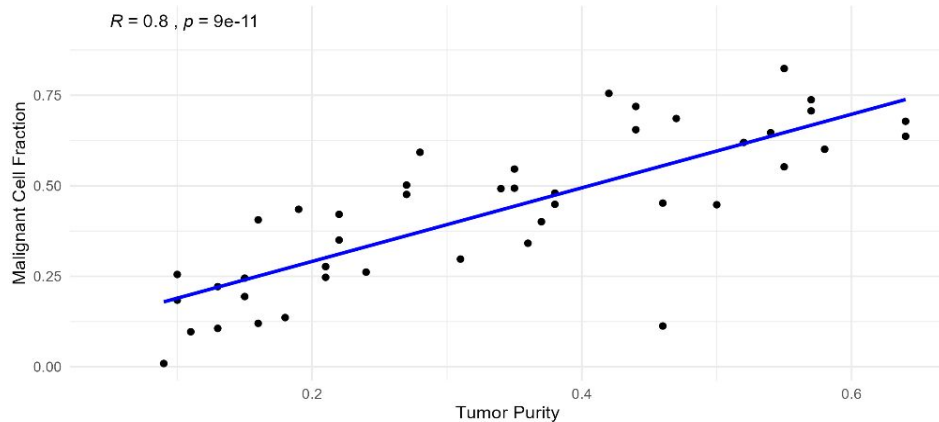
EcoTyper, a machine learning framework for the identification of cell states and ecosystems from bulk, single-cell, and spatially-resolved expression data, is now available. EcoTyper extends CIBERSORTx for large-scale profiling of cellular ecosystems.

The CIBERSORT legacy website has been incorporated into the CIBERSORTx website under the menu item **CS Archives**.

Correlating deconvolution results with tumor purity

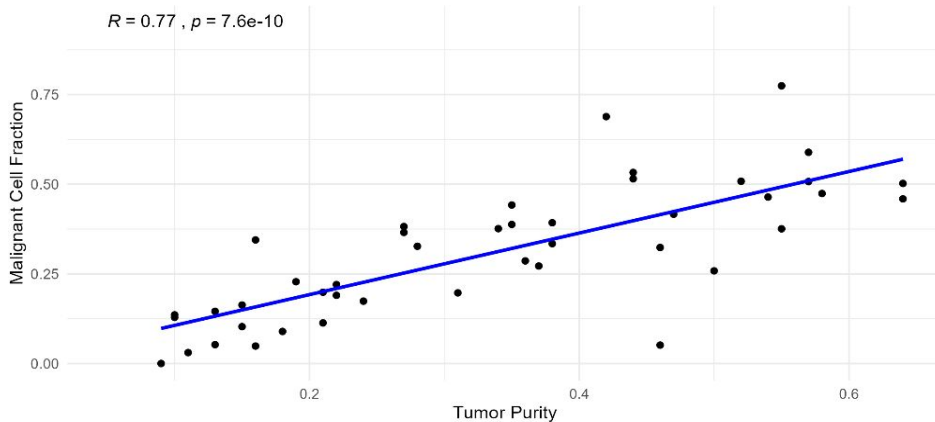
Correlation between Tumor Purity and Malignant Cell Fraction - BayesPrism

$R = 0.8$, $p = 9e-11$



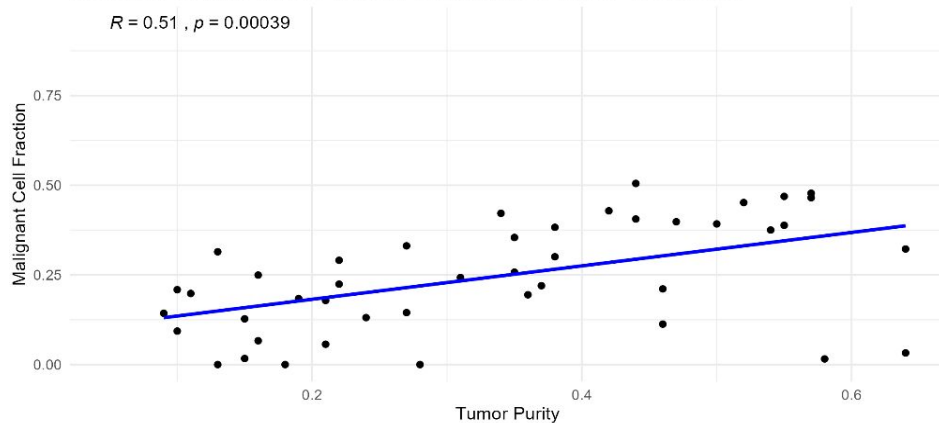
Correlation between Tumor Purity and Malignant Cell Fraction - BayesPrism (Top25 markers)

$R = 0.77$, $p = 7.6e-10$



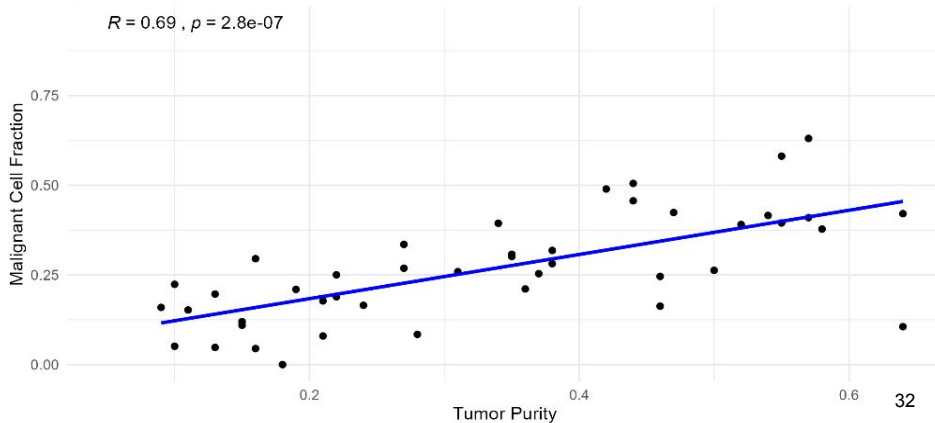
Correlation between Tumor Purity and Malignant Cell Fraction - BisqueRNA

$R = 0.51$, $p = 0.00039$



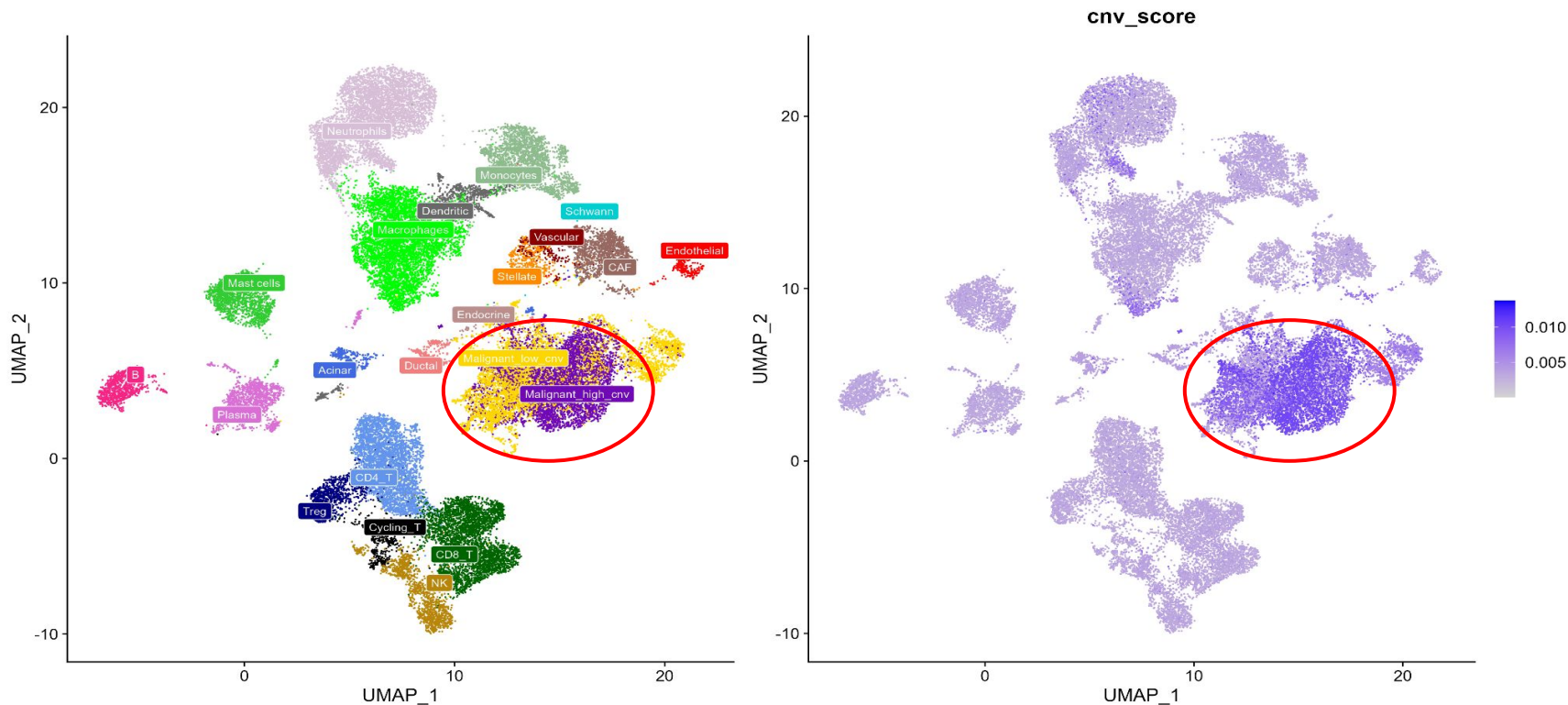
Correlation between Tumor Purity and Malignant Cell Fraction - BisqueRNA (Top25 markers)

$R = 0.69$, $p = 2.8e-07$



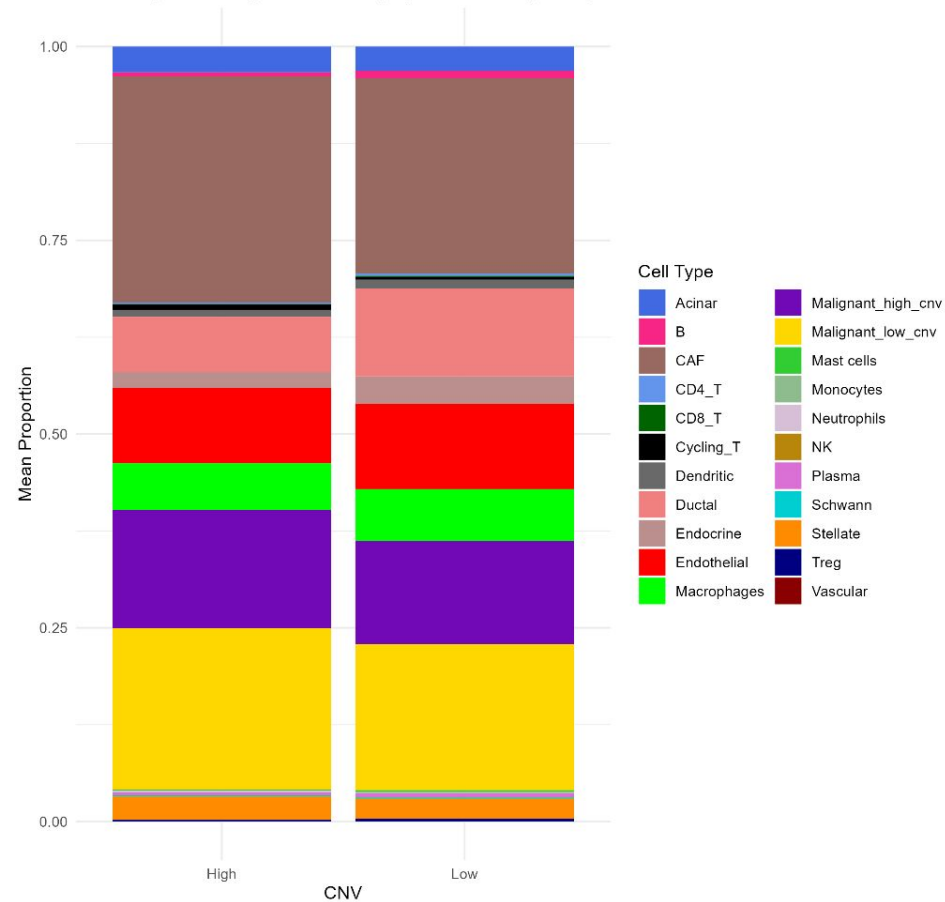
Correlating deconvolution results with CNV

Do samples with higher malignant cell proportions (from deconvolution) show distinct CNV patterns?

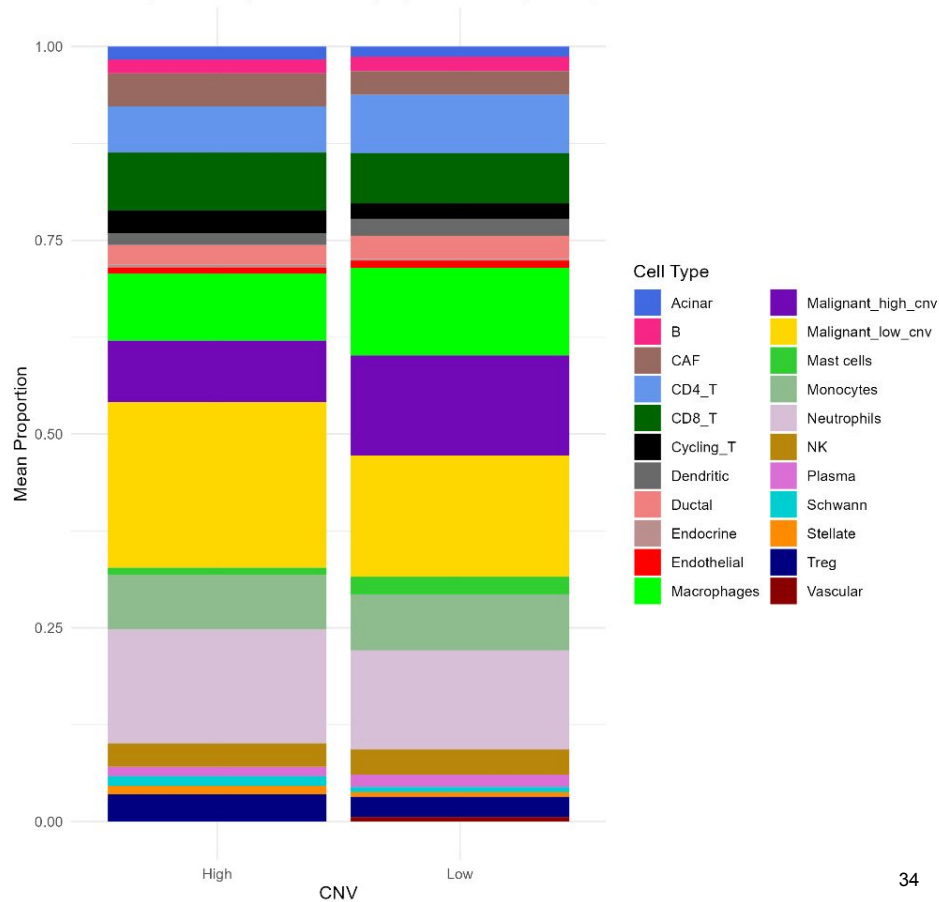


Correlating deconvolution results with CNV

Cell Proportions by CNV class (Top25 markers) - BayesPrism

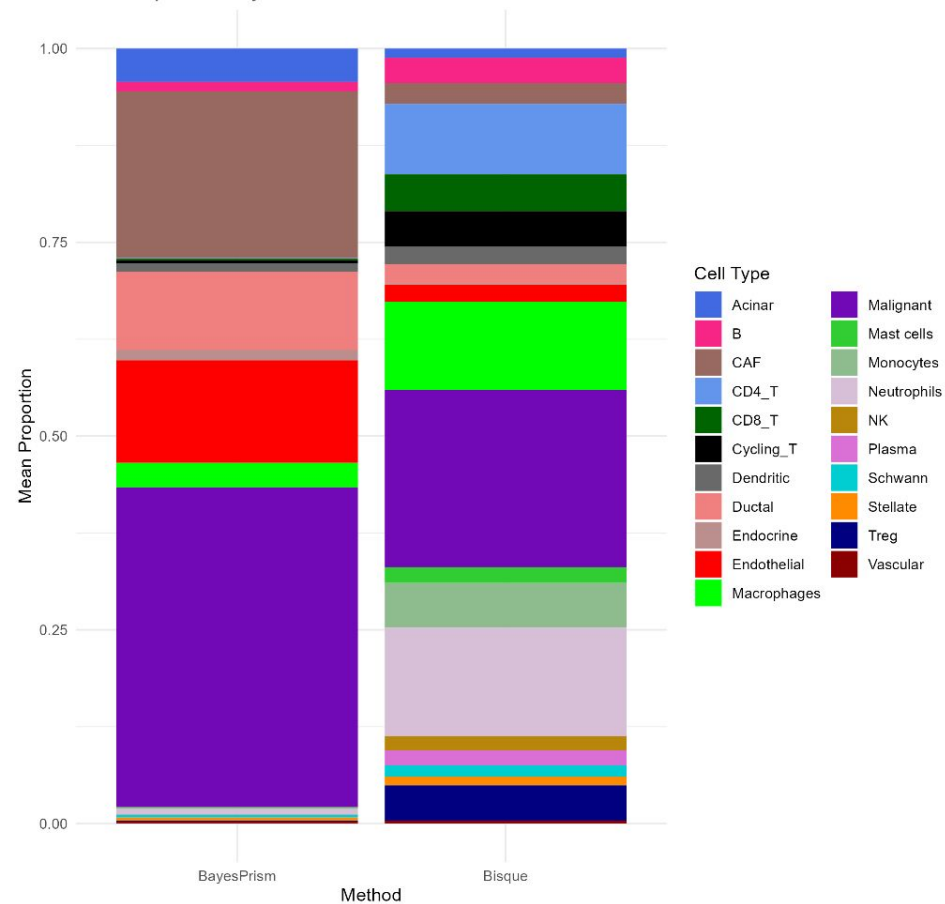


Cell Proportions by CNV class (Top25 markers) - BisqueRNA

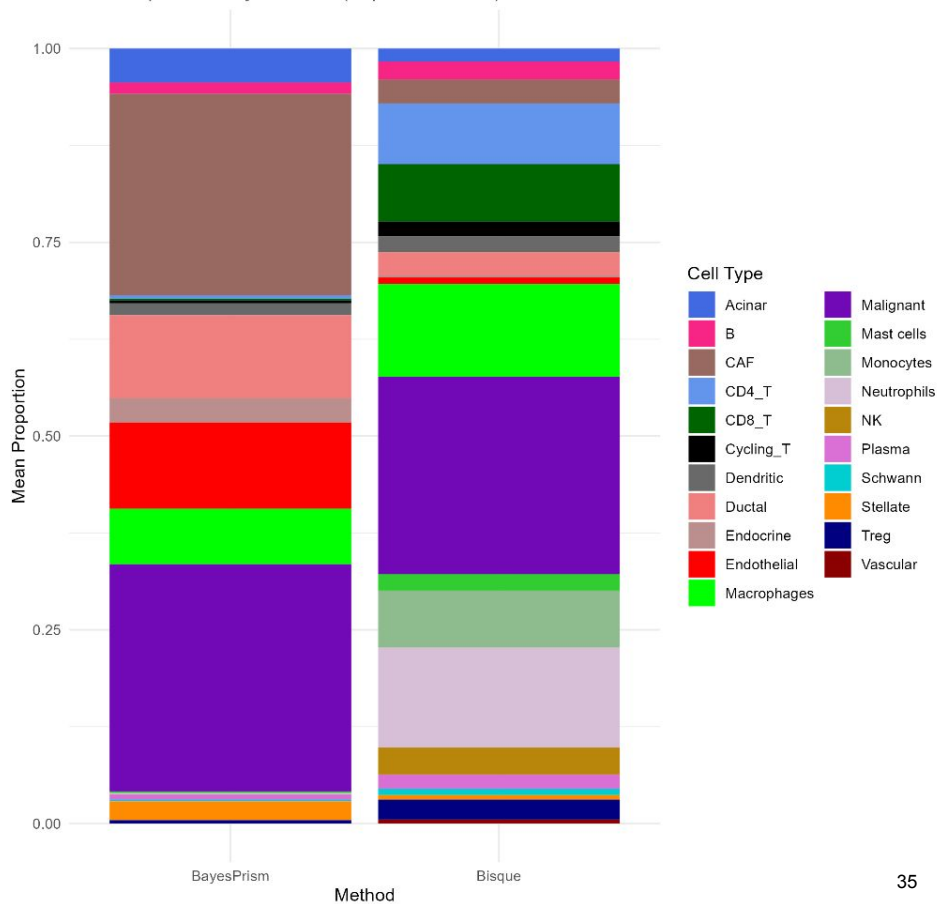


Comparing deconvolution results between methods

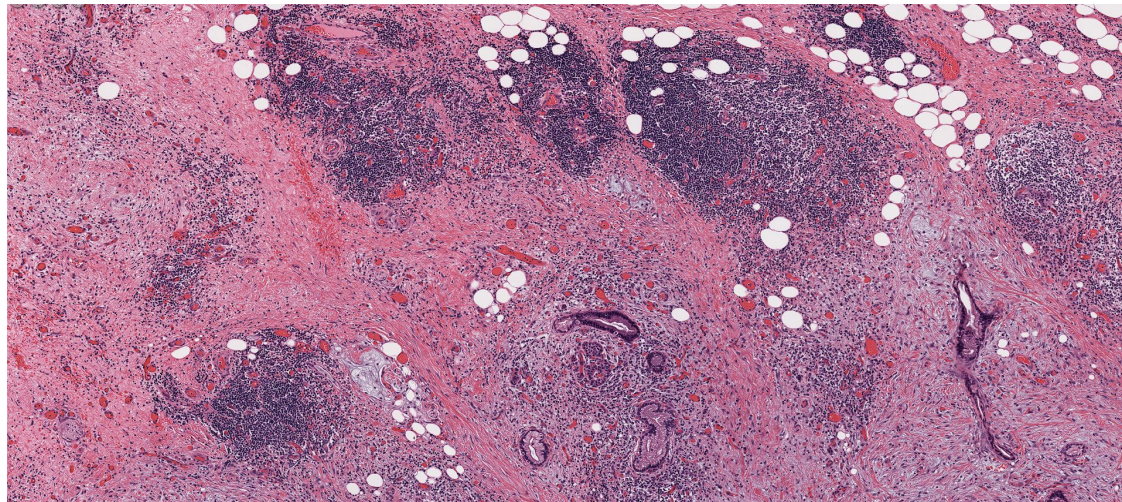
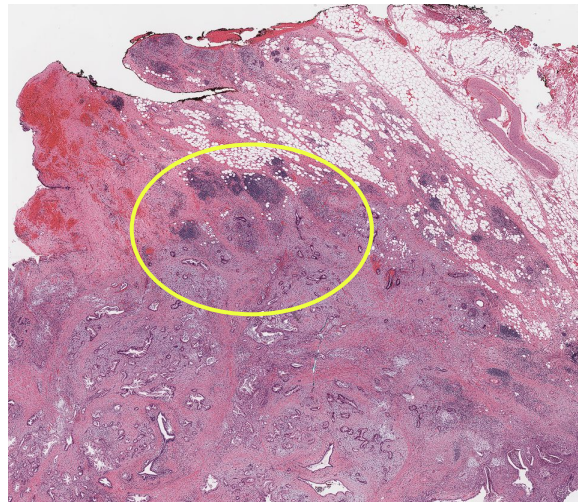
Cell Proportions by Method



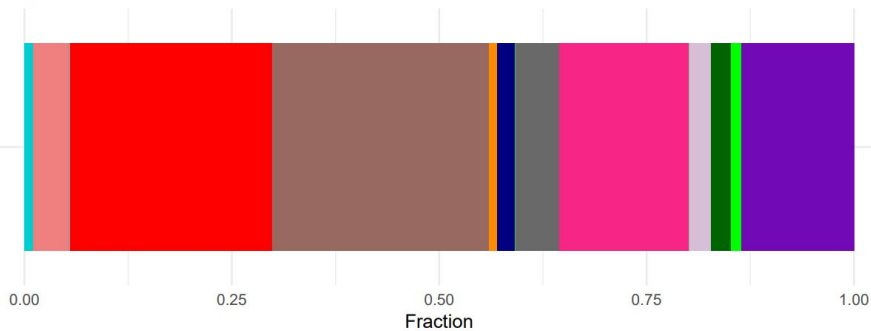
Cell Proportions by Method (Top25 markers)



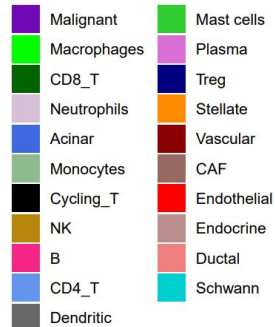
Correlating deconvolution results with H&E slides



Composition of TCGA-IB-AAUR-01A-21R-A38C-07



variable



Challenge

Present one research question and conduct a downstream analysis using your deconvolution results.

- Cluster bulk samples
- Correlate with clinical covariates
- Correlate deconvolution results with tumor purity
- Z-scores of signature genes of the cell type of interest
- Survival analysis
- How gene expression in malignant cells correlates with the cell type fraction of non-malignant cells

Exercise 4

Final Quiz

- . How can I validate my deconvolution results?
- . What is the spillover effect?
- . How can I deal with rare cell types or cell states?
- . What level of cell annotation (resolution) should I use?
- . How can deconvolution methods be improved?



Find more resources at the github repository

<https://github.com/estefvazquez/scDeconvolution>

Acknowledgments

EMBL EBI Training Team

Cancer Genetics & Bioinformatics Lab - LIIGH UNAM

Immunotherapy Lab UFCSPA

EMBL-EBI

